

## Vpc Task -1

1.Create VPC with 2 private and 2 public subnets.

Steps to create a vpc :

- 1) Search for vpc
- 2) Open vpc
- 3) Click on create vpc

The screenshot shows the 'Create VPC' wizard in the AWS Management Console, specifically the 'VPC settings' step. The 'Resources to create' section has 'VPC and more' selected. Under 'Name tag auto-generation', 'Auto-generate' is checked, and the name 'Practice-vpc-with-subnets' is entered. The 'IPv4 CIDR block' is set to '10.0.0.0/16', which provides 65,536 IP addresses. The 'IPv6 CIDR block' is set to 'No IPv6 CIDR block'. The 'Tenancy' is set to 'Default'. The 'Number of Availability Zones (AZs)' is set to 2. A 'Preview' section on the right shows a diagram of the VPC structure: a central VPC box labeled 'Practice-vpc-with-subnets-vpc' is connected to four subnets: 'eu-north-1a' and 'eu-north-1b' (both private), and two public subnets. These public subnets are each connected to a route table, which in turn connects to an internet gateway (represented by a cloud icon).

The screenshot shows the 'Subnet settings' step of the 'Create VPC' wizard. It is for 'Subnet 1 of 1'. The 'Subnet name' is 'Public subnet-1'. The 'Availability Zone' is 'Europe (Stockholm) / eu-north-1a'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.8.0/23', which provides 512 IP addresses. The 'Tags' section shows a key-value pair: 'Name' with the value 'Public subnet-1'. There is an 'Add new tag' button and a 'Remove' button for the existing tag.

ubnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public subnet-2

The name can be up to 256 characters long.

Availability Zone

[Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Europe (Stockholm) / eun1-az1 (eu-north-1a)

IPv4 VPC CIDR block

[Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.10.0/23 512 IPs

< > ^ v

▼ Tags - optional

Key

Q Name X

Value - optional

Q Public subnet-2 X Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Privatesubnet-1

The name can be up to 256 characters long.

Availability Zone

[Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Europe (Stockholm) / eun1-az1 (eu-north-1a)

IPv4 VPC CIDR block

[Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.10.0/23

512 IPs

▼ Tags - optional

Key

Q Name X

Value - optional

Q Privatesubnet-1 X Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Privatesubnet-2

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Europe (Stockholm) / eun1-az1 (eu-north-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.10.0/23

512 IPs

▼ Tags - optional

Key

Q Name



Value - optional

Q Privatesubnet-2



Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

## 2.Enable DNS Hostname in VPC.

### Edit VPC settings [Info](#)

#### VPC details

VPC ID

vpc-02c59801748cb627b

Name

Pratice-vpc-with-subnets -vpc

#### DHCP settings

DHCP option set [Info](#)

dopt-010243c41ee0ebba1

#### DNS settings

☒ Enable DNS resolution [Info](#)

☒ Enable DNS hostnames [Info](#)

#### Network Address Usage metrics settings

☐ Enable Network Address Usage metrics [Info](#)

### 3.Enable Auto Assign Public IP in 2 public subnets.

**Edit subnet settings** [Info](#)

**Subnet**

Subnet ID  
subnet-06f0112df2f6e270f

Name  
Pratice-vpc-with subnet 1 with public1-eu-north-1b

**Auto-assign IP settings** [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)  
Option disabled because no customer owned pools found.

**Resource-based name (RBN) settings** [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

☐ Enable resource name DNS AAAA record on launch [Info](#)

**Hostname type** [Info](#)

☐ Resource name

☒ IP name

**DNS64 settings**

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

☐ Enable DNS64 [Info](#)

[Cancel](#) [Save](#)

**Edit subnet settings** [Info](#)

**Subnet**

Subnet ID  
subnet-00f22a9e3d61c12fb

Name  
Pratice-vpc-with-subnet2 public2-eu-north-1b

**Auto-assign IP settings** [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)  
Option disabled because no customer owned pools found.

**Resource-based name (RBN) settings** [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

☐ Enable resource name DNS AAAA record on launch [Info](#)

**Hostname type** [Info](#)

☐ Resource name

☒ IP name

**DNS64 settings**

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

☐ Enable DNS64 [Info](#)

### 4.Add 2 private subnets in private route table.

**Route tables (1/9)** [Info](#)

Last updated 23 minutes ago [Actions](#)

Find route tables by attribute or tag

| <input type="checkbox"/>            | Name                                       | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                               | Owner ID     |
|-------------------------------------|--------------------------------------------|-----------------------|---------------------------|-------------------|------|-----------------------------------|--------------|
| <input checked="" type="checkbox"/> | Pratice-vpc-with-subnets -rtb-private1-... | rtb-01a2248d7de80ae05 | subnet-0f947f1b291e21...  | -                 | No   | vpc-0b54a94f10c6c41af   Pratic... | 814787354428 |
| <input type="checkbox"/>            | -                                          | rtb-0746c57bcffb99236 | -                         | -                 | Yes  | vpc-00ce773080d81479f             | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-private2-... | rtb-0788b98eaf3afb1c4 | subnet-0a06e2789664db...  | -                 | No   | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-public       | rtb-085f29533184155ea | 2 subnets                 | -                 | No   | vpc-0b54a94f10c6c41af   Pratic... | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-public       | rtb-07f3007e3e5f9efda | 2 subnets                 | -                 | No   | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-private1-... | rtb-01f2a853fb4234503 | subnet-0f819a1a49f03b...  | -                 | No   | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-private2-... | rtb-07be33ced6b8912ef | subnet-075b8755f07f6c...  | -                 | No   | vpc-0b54a94f10c6c41af   Pratic... | 814787354428 |
| <input type="checkbox"/>            | -                                          | rtb-0e512c75881434a62 | -                         | -                 | Yes  | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input type="checkbox"/>            | -                                          | rtb-0d04b95087e6e7bfc | -                         | -                 | Yes  | vpc-0b54a94f10c6c41af   Pratic... | 814787354428 |

## 5.Add 2 public subnets in public route table.

|                                     |                                            |                       |                          |   |     |                                   |              |
|-------------------------------------|--------------------------------------------|-----------------------|--------------------------|---|-----|-----------------------------------|--------------|
| <input type="checkbox"/>            | -                                          | rtb-0746c57bcffb99236 | -                        | - | Yes | vpc-00ce773080d81479f             | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-private2-... | rtb-0788b98eaf3afb1c4 | subnet-0a06e2789664db... | - | No  | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input checked="" type="checkbox"/> | Pratice-vpc-with-subnets -rtb-public       | rtb-085f29533184155ea | 2 subnets                | - | No  | vpc-0b54a94f10c6c41af   Pratic... | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-public       | rtb-07f3007e3e5f9efda | 2 subnets                | - | No  | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-private1-... | rtb-01f2a853fb4234503 | subnet-0f819a1a49f03b... | - | No  | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input type="checkbox"/>            | Pratice-vpc-with-subnets -rtb-private2-... | rtb-07be33ced6b8912ef | subnet-075b8755f07f6c... | - | No  | vpc-0b54a94f10c6c41af   Pratic... | 814787354428 |
| <input type="checkbox"/>            | -                                          | rtb-0e512c75881434a62 | -                        | - | Yes | vpc-02c59801748cb627b   Prat...   | 814787354428 |
| <input type="checkbox"/>            | -                                          | rtb-0d04b95087e6e7bfc | -                        | - | Yes | vpc-0b54a94f10c6c41af   Pratic... | 814787354428 |

### rtb-085f29533184155ea / Pratice-vpc-with-subnets -rtb-public

Details **Routes** Subnet associations Edge associations Route propagation Tags Related resources

**Routes (2)**

Filter routes

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | igw-0e64a2a59124ba103 | Active | No         | Create Route       |
| 10.0.0.0/16 | local                 | Active | No         | Create Route Table |

## 6.Public route table will have the routes to internet and local.

Route table rtb-08d12948e72efb6ea | route table public was created successfully.

**Route tables (1/2)** [Info](#)

Last updated 1 minute ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

| <input type="checkbox"/>            | Name               | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                   | Owner ID     |
|-------------------------------------|--------------------|-----------------------|---------------------------|-------------------|------|-----------------------|--------------|
| <input type="checkbox"/>            | -                  | rtb-0746c57bcffb99236 | -                         | -                 | Yes  | vpc-00ce773080d81479f | 814787354428 |
| <input checked="" type="checkbox"/> | route table public | rtb-08d12948e72efb6ea | -                         | -                 | No   | vpc-00ce773080d81479f | 814787354428 |

## 7.Create EC2 in public subnet with t2.micro and install PHP.

```
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.5-1.amzn2023.0.1.x86_64          apr-util-ldap-1.6.5-1.amzn2023.0.1.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.68-1.amzn2023.0.1.x86_64          httpd-core-2.4.68-1.amzn2023.0.1.x86_64
httpd-tools-2.4.68-1.amzn2023.0.1.x86_64          libbrotli-1.1.0-1.amzn2023.0.1.x86_64          libsodium-1.0.19-5.amzn2023.x86_64
mod_http2-2.0.42-1.amzn2023.0.1.x86_64          mod_lua-2.4.68-1.amzn2023.0.1.x86_64          nginxfilesystem-1:1.30.5-1.amzn2023.0.1.noarch
php8.5-cli-8.5.10-1.amzn2023.0.1.x86_64          php8.5-common-8.5.10-1.amzn2023.0.1.x86_64    php8.5-fpm-8.5.10-1.amzn2023.0.1.x86_64
php8.5-mysqlnd-8.5.10-1.amzn2023.0.1.x86_64      php8.5-pdo-8.5.10-1.amzn2023.0.1.x86_64      php8.5-process-8.5.10-1.amzn2023.0.1.x86_64
php8.5-xml-8.5.10-1.amzn2023.0.1.x86_64

Complete!

[ec2-user@ip-172-31-35-137 html]$ php -v
PHP 8.5.10 (cli) (built: Aug 25 2026 21:09:32) (NTS gcc x86_64)
Copyright (c) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.10, Copyright (c) Zend Technologies
with Zend OPcache v8.5.10, Copyright (c), by Zend Technologies
```

▼ Network settings

Info

VPC - required

Info

vpc-0d972d117f298591c (PHP-VPC)

10.0.0.0/16

Subnet

Info

subnet-0355d4bee869790e9

PHP subnet

VPC: vpc-0d972d117f298591c Owner: 814787354428 Availability Zone: eu-north-1a (eun1-az1)  
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Create new subnet

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups

Info

Select security groups

Compare security group rules

PHP-Security group sg-06aec9cee516a22bd

VPC: vpc-0d972d117f298591c

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

| PHP Version 8.5.10                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| System                                                                                                                                                                                           | Linux ip-10-0-1-117.eu-north-1.compute.internal 6.18.51-120.162.amzn2023.x86_64 #2 SMP PREEMPT_DYNAMIC Wed Sep 23 18:50:41 UTC 2026 x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Build Date                                                                                                                                                                                       | Aug 25 2026 21:09:32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Build System                                                                                                                                                                                     | Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Build Provider                                                                                                                                                                                   | Amazon Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Compiler                                                                                                                                                                                         | gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Architecture                                                                                                                                                                                     | x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Server API                                                                                                                                                                                       | FPM/FastCGI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Virtual Directory Support                                                                                                                                                                        | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Configuration File (php.ini) Path                                                                                                                                                                | /etc                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Loaded Configuration File                                                                                                                                                                        | /etc/php.ini                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Scan this dir for additional .ini files                                                                                                                                                          | /etc/php.d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Additional .ini files parsed                                                                                                                                                                     | /etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mysqld.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-posix.ini, /etc/php.d/20-shmop.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sodium.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-sysmsg.ini, /etc/php.d/20-sysvsem.ini, /etc/php.d/20-sysvshm.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-mysqli.ini, /etc/php.d/30-pdo_mysql.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmldr.ini |
| PHP API                                                                                                                                                                                          | 20250925                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| PHP Extension                                                                                                                                                                                    | 20250925                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Zend Extension                                                                                                                                                                                   | 420250925                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Zend Extension Build                                                                                                                                                                             | API420250925.NTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| PHP Extension Build                                                                                                                                                                              | API20250925.NTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| PHP Integer Size                                                                                                                                                                                 | 64 bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Debug Build                                                                                                                                                                                      | no                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Thread Safety                                                                                                                                                                                    | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Zend Signal Handling                                                                                                                                                                             | enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Zend Memory Manager                                                                                                                                                                              | enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Zend Multibyte Support                                                                                                                                                                           | provided by mbstring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Zend Max Execution Timers                                                                                                                                                                        | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| IPv6 Support                                                                                                                                                                                     | enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DTrace Support                                                                                                                                                                                   | available, disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Registered PHP Streams                                                                                                                                                                           | https, ftps, compress.zlib, php, file, glob, data, http, ftp, compress.bzip2, phar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Registered Stream Socket Transports                                                                                                                                                              | tcp, udp, unix, udg, ssl, tls, tlsv1.0, tlsv1.1, tlsv1.2, tlsv1.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Registered Stream Filters                                                                                                                                                                        | zlib.*, string.rot13, string.toupper, string.tolower, convert.*, consumed, dechunk, bzip2.*, convert.iconv.*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| This program makes use of the Zend Scripting Language Engine:<br>Zend Engine v4.4.0, Copyright (c) 2025 by Zend Technologies, with Zend OPcache v8.5.10, Copyright (c) 2025 by Zend Technologies |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

8.Configure NAT gateway in public subnet and connect to private instance.

- a) Create vpc
- b) VPC → Subnets → Create subnet
- c) VPC → Subnets → Create subnet
- d) VPC → Internet Gateways→ Click on Create internet gateway
- e) Select nat-igw
- f) Click on Actions → Attach to a vpc
- g) Select nat-vpc
- h) Click on Attach internet gateway
- I)Create nat gate way

## nat-19e9511c2e6391cd6 / Nat-gateway-01

Actions ▾

## Details

## NAT gateway ID

nat-19e9511c2e6391cd6

## NAT gateway ARN

arn:aws:ec2:eu-north-1:814787354428:natgateway/nat-19e9511c2e6391cd6

## VPC

vpc-0a4db82daa7638079 / Nat-Vpc

## Availability mode

Regional

## Connectivity type

Public

## Method of EIP allocation

Automatic

## State

Available

## Created

Tuesday, September 29, 2026 at 22:39:21 GMT+5:30

## State message

-

## Deleted

-

IP addresses

Monitoring

Flow logs

Tags

## Associated IP addresses (1)



Edit IP address associations

Q Search

&lt; 1 &gt; ⚙

| IP address    | Status      | Availability Zone      | Allocation ID              | Ass  |
|---------------|-------------|------------------------|----------------------------|------|
| 16.170.17.217 | ✓ Succeeded | eun1-az3 (eu-north-1c) | eipalloc-0e3e37332206523e1 | eipa |

```
[ec2-user@ip-10-0-2-244 ~]$ ping google.com
PING google.com (142.251.142.238) 56(84) bytes of data.
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=1 ttl=119 time=3.71 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=2 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=3 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=4 ttl=119 time=3.79 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=5 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=6 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=7 ttl=119 time=3.84 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=8 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=9 ttl=119 time=3.78 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=10 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=11 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=12 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=13 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=14 ttl=119 time=3.83 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=15 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=16 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=17 ttl=119 time=3.73 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=18 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=19 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=20 ttl=119 time=3.77 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=21 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=22 ttl=119 time=3.76 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=23 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=24 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=25 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=26 ttl=119 time=3.80 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=27 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=28 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=29 ttl=119 time=3.82 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=30 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=31 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=32 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=33 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=34 ttl=119 time=3.78 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=35 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=36 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=37 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=38 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=39 ttl=119 time=3.73 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=40 ttl=119 time=3.73 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=41 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=42 ttl=119 time=3.80 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=43 ttl=119 time=3.74 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=44 ttl=119 time=3.75 ms
64 bytes from lcarnb-ah-in-f14.1e100.net (142.251.142.238): icmp_seq=45 ttl=119 time=3.76 ms
```

9.Install Apache Tomcat in private EC2 and deploy a sample app.

```
Installed:
alsa-lib-1.2.7-2.1.amzn2023.0.4.x86_64
dejavu-sans-mono-fonts-2.37-16.amzn2023.0.2.noarch
fonts-filesystem-1:2.0.5-12.amzn2023.0.2.noarch
google-noto-fonts-common-20240401-1.amzn2023.0.2.noarch
harfbuzz-7.0.0-2.amzn2023.0.2.x86_64
javadoc-filesystem-6.0.0-7.amzn2023.0.6.noarch
libSM-1.2.4-3.amzn2023.0.1.x86_64
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
cairo-1.18.0-4.amzn2023.0.3.x86_64
dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch
freetype-2.13.2-5.amzn2023.0.2.x86_64
google-noto-sans-vf-fonts-20240401-1.amzn2023.0.2.noarch
java-17-amazon-corretto-1:17.0.20-10-1.amzn2023.1.x86_64
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libX11-1.8.10-2.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libpng-2:1.6.27-10.amzn2023.0.13.x86_64
xsl-common-0.6.3-56.amzn2023.0.2.noarch
dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
glib-3.2.11-9.amzn2023.0.4.x86_64
graphite2-1.3.14-7.amzn2023.0.3.x86_64
java-17-amazon-corretto-headless-1:17.0.20-10-1.amzn2023.1.x86_64
libICE-1.1.1-1.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libX11-1.8.2-1.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libbrotli-1.1.0-1.amzn2023.0.1.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-0-241 ~]$ java -version
openjdk version "17.0.20.1" 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-17.0.20.10-1 (build 17.0.20.1-10-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.20.10-1 (build 17.0.20.1-10-LTS, mixed mode, sharing)

[ec2-user@ip-172-31-0-241 bin]$ curl -I https://google.com
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-Z2E-gR
p.withgoogle.com/csp/gws/other-hp
date: Wed, 30 Sep 2026 10:35:24 GMT
expires: Fri, 30 Oct 2026 10:35:24 GMT
cache-control: public, max-age=2592000
server: gws
content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
```

## 10.Configure VPC flow logs and store the logs in S3 and CloudWatch.

Log group "Vpc-flowlogs" has been created.

Log groups

Data sources - new

Summary

Get-set-go with unified data store

Log groups (1)

By default, we only load up to 10,000 log groups.

☒ Exact match

< 1 >

| <input type="checkbox"/> | Log group    | Log class | Anomaly d...              | Deletion protec...        | Bearer token a...         | Data protecti... | Sensitive dat... | Retention    | Metric filters | Contribi |
|--------------------------|--------------|-----------|---------------------------|---------------------------|---------------------------|------------------|------------------|--------------|----------------|----------|
| <input type="checkbox"/> | Vpc-flowlogs | Standard  | <a href="#">Configure</a> | <input type="radio"/> Off | <input type="radio"/> Off | -                | -                | Never expire | -              | :        |

Roles (1/4) [info](#)

Delete

Create role

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

< 1 >

| <input checked="" type="checkbox"/> | Role name                                         | Trusted entities                                       | Last activity |
|-------------------------------------|---------------------------------------------------|--------------------------------------------------------|---------------|
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForResourceExplorer</a> | AWS Service: resource-explorer-2 (Service-Linked Role) | 1 hour ago    |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForSupport</a>          | AWS Service: support (Service-Linked Role)             | -             |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForTrustedAdvisor</a>   | AWS Service: trustedadvisor (Service-Linked Role)      | -             |
| <input checked="" type="checkbox"/> | VPCflowlogrole                                    | AWS Service: ec2                                       | -             |

▼ Log group details

Log class [Info](#)

Standard

ARN  
 arn:aws:logs:eu-north-1:814787354428:log-group:Vpc-flowlogs\*

Creation time  
-

Retention  
Never expire

Stored bytes  
-

Account  
 -

Metric filters

Subscription filters  
Not supported

Contributor Insights rules  
Not supported

KMS key ID  
-

Deletion protection  
 Off

Data protection  
-

Sensitive data count  
Not supported

Custom field indexes  
Not supported

Transformer  
Not supported

Anomaly detection  
Not supported

Bearer token authentication  
 Off

Syslog ingestion [Info](#)  
[Configure](#)

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Field indexes

Transformer

Log streams (0)

[Delete](#) [Create log stream](#) [Search all log streams](#)

By default, we only load the most recent log streams.

☐ Exact match ☐ Show expired [Info](#)

< 1 >

Log stream

▼ Last event time

▼

There are no log streams.

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1/1) [Info](#)

[Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

< 1 >

Name ▲

AWS Region ▼

Creation date ▼

[my-vpc-flow-logs-2026-abhishek](#)

Europe (Stockholm) eu-north-1

September 30, 2026, 16:19:07 (UTC+05:30)